

Ibrahim Mohamed

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EDUCATION

Ain Shams University

Bachelor of Computer Science

Cairo, Egypt

June 2026 (*Expected*)

- **Concentration:** Computer Systems (Distributed Systems, Cloud Computing, Backend Development).
- **Courses:** Distributed Systems, Operating Systems, Networking, Design Pattern, Data Structures & Algorithms, Computer Architecture, OOP, Database Systems, Artificial Intelligence, Machine Learning, Computer Vision.

EXPERIENCE

Founding Software Engineer

Pirates Company · Cairo, Egypt

Oct. 2025 – Present

[Live Storefront](#)

- End-to-end owner of a production-oriented ERP and e-commerce platform for a sauce company, responsible for architecture, backend, frontend, infrastructure, deployment, observability, and operational reliability.
- Built the backend system of record with FastAPI, PostgreSQL, Redis, and Celery, implementing business-critical workflows for orders, vendor management, production, payments, credit notes, and document/report generation.
- Implemented transactional order and finance logic connecting FIFO inventory deduction, stored customer/vendor balances, double-entry journal entries, and reversal-based cancellations, with row-level locking and idempotent boundaries for consistency and auditability.
- Built the internal admin dashboard and public storefront workflows, including catalog browsing, guest checkout, payment integration, order tracking, approval flows, and operational tooling for internal users.
- Designed and deployed the production stack on AWS using Terraform, GitHub Actions, EC2, RDS, S3, ECR, Nginx, and CloudWatch, including infrastructure automation, CI/CD, migrations, and service rollout workflows.
- Built and continuously improved production safeguards and observability through Prometheus, Grafana, rate limiting, webhook idempotency, background job processing, and monitoring/alerting to strengthen reliability and operational visibility.

PROJECTS

Distributed Key-Value Store | Go, gRPC, TCP | [GitHub](#)

Dec. 2025 – Jan. 2026

- Implemented fault-tolerant KV store with Raft consensus: leader election, log matching, and linearizable writes.
- Built log replication across 3-node cluster with crash recovery and network partition handling.
- Added log compaction via snapshots to bound storage growth.

Storage Engine | Rust, LSM-Tree | [GitHub](#)

Feb. 2026 – Mar. 2026

- Built LSM-tree key-value engine from scratch: memtable, WAL, SSTables, compaction.
- Implemented bloom filters, block cache (LRU), and k-way merge iterators.
- Added crash recovery via WAL replay and manifest-based version tracking.

API Gateway | Go, Prometheus | [GitHub](#)

Jan. 2026 – Mar. 2026

- Built production-grade API gateway from scratch with zero framework dependencies.
- Implemented 4 load balancing strategies, 3 rate limiting algorithms, and circuit breakers.
- Added active/passive health checks, YAML hot-reload routing, and Prometheus metrics.

Smart Cloud Optimizer | Python, AWS, Prophet, OR-Tools | [GitHub](#)

Sep. 2025 - June. 2026

- Built AI module recommending AWS setups for new users via structured prompts to LLMs.
- Implemented ML forecasting (Prophet/SARIMAX) and cost optimization (OR-Tools/PuLP).
- Integrated AWS APIs (Cost Explorer, CloudWatch, Pricing) to analyze usage and cut costs.

TECHNICAL SKILLS

Programming Languages: Python, Go, TypeScript, SQL, Rust, C/C++.

Databases: PostgreSQL, MySQL, Redis, MongoDB, Firebase.

Infrastructure: AWS, Docker, Kubernetes, Terraform, Kafka, Prometheus, Grafana, Linux, Bash, Git, GitHub Actions.

Frameworks: FastAPI, Flask, Gin, NestJS, Next.js, Axum, React, gRPC.

Concepts: System Design, Microservices, Stream Processing, Observability, SRE, CI/CD, Testing, SOLID Principles.

Languages: Arabic (Native), English (Intermediate).